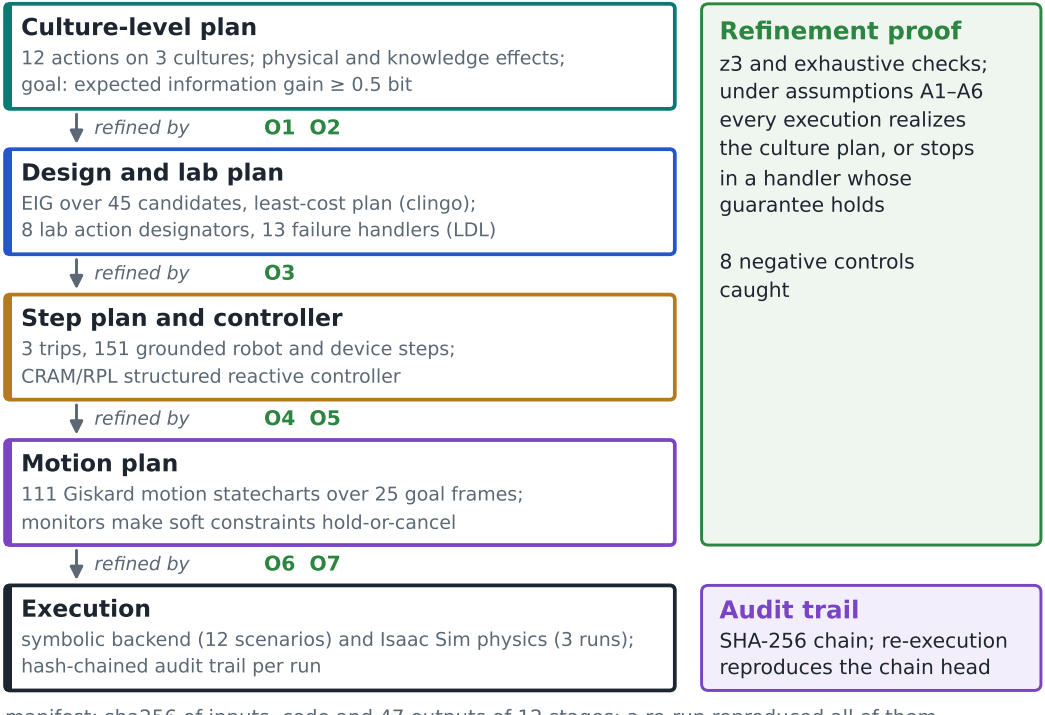
Culture-level plan
12 actions on 3 cultures; physical and knowledge effects;
goal: expected information gain ≥ 0.5 bit
refined by
O1 O2
Design and lab plan
EIG over 45 candidates, least-cost plan (clingo);
8 lab action designators, 13 failure handlers (LDL)
refined by
O3
Step plan and controller
3 trips, 151 grounded robot and device steps;
CRAM/RPL structured reactive controller
refined by
O4 O5
Motion plan
111 Giskard motion statecharts over 25 goal frames;
monitors make soft constraints hold-or-cancel
refined by
O6 O7
Execution
symbolic backend (12 scenarios) and Isaac Sim physics (3 runs);
hash-chained audit trail per run
Refinement proof
z3 and exhaustive checks;
under assumptions A1–A6
every execution realizes
the culture plan, or stops
in a handler whose
guarantee holds
8 negative controls
caught
Audit trail
SHA-256 chain; re-execution
reproduces the chain head
manifest: sha256 of inputs, code and 47 outputs of 12 stages; a re-run reproduced all of them